

FakeQuantize#

https://pytorch.org/docs/stable/generated/torch.nn.quantization.fake_quantize.html

Description:

Simulate the quantize and dequantize operations in training time. The output of this module is given by: `is_dynamic` indicates whether the fake quantize is a placeholder for dynamic quantization operators (`choose_qparams -> q -> dq`) or static quantization operators (`q -> dq`). `scale` defines the scale factor used for quantization. `zero_point` specifies the quantized value to which 0 in floating point maps to.

Function Signature

```
class
torch.ao.quantization.fake_quantize.FakeQuantize(observer=
<class
'torch.ao.quantization.observer.MovingAverageMinMaxObserver'
>, quant_min=None, quant_max=None, is_dynamic=False,
**observer_kwargs)[source]#
```

dtype specifies the quantized dtype that is being emulated with fake-quantization,

Parameters

Variables

Code Examples

```
x_out = (
    clamp(round(x / scale + zero_point), quant_min, quant_max) -
    zero_point
) * scale
```

Detailed Documentation

Documentation

Simulate the quantize and dequantize operations in training time.

The output of this module is given by:

`is_dynamic` indicates whether the fake quantie is a placeholder for dynamic quantization operators (choose_qparams -> q -> dq) or static quantization operators (q -> dq)

`scale` defines the scale factor used for quantization.

`zero_point` specifies the quantized value to which 0 in floating point maps to

`fake_quant_enabled` controls the application of fake quantization on tensors, note that statistics can still be updated.

`observer_enabled` controls statistics collection on tensors

allowable values are `torch.qint8` and `torch.quint8`.

`observer (module)` – Module for observing statistics on input tensors and calculating scale and zero-point.

`observer_kwargs (optional)` – Arguments for the observer module

`activation_post_process (Module)` – User provided module that collects statistics on the input tensor and provides a method to calculate scale and zero-point.

